



TLD SERVICES

Terminology and Definitions

Index:

1	Objectives:.....	3
2	Service terminology and definitions	3
2.1	Site activities:.....	3
	• Equipment Warranty Period Support:	3
	• Technical Assistance:	3
	• Troubleshooting and service calls:.....	3
	• Audit and Expertise:.....	3
	• WC, SB:.....	3
	• Training.....	4
	• TKMC.....	4
2.2	Spare parts:	4
	• Capital spares.....	4
	• Spare parts:	4
3	Maintenance terminology and definition:	4
3.1	Preventive Maintenance (PM/PMI)	4
3.2	Corrective Maintenance (CM)	5
3.3	Major service:	5
3.4	Repair:	5
3.5	Overhaul:	5
3.6	Refurbishment:	Erreur ! Signet non défini.
3.7	Modernization:	5
4	Maintainability Performance Criteria:	6
4.1	Operating Time:	6
4.2	Reliability, maintainability, availability, MTBF and MTTR.....	7
	Reliability (MTBF):	7
	Maintainability (MTTR)	8
	Availability.....	8
5	Total Cost of Ownership (TCO):	8
5.1	Introduction:	9
5.2	Objective:.....	9
5.3	Terminology	9
	5.3.1 Cost drivers:	9
	5.3.2 Timeframe:	11
	5.3.3 Unit of value:.....	11
	5.3.4 Which TCO?	11
	5.3.5 Model:	11

1 **Objectives:**

The objective of this document is to set a common language across TLD group in terms of customer services. It sets out the different terms and their appropriate definitions, maintainability performance criteria and finally, Total Cost of Ownership (Life Cycle methodology).

This document is part of the TLD Group procedures

2 **Service terminology and definitions**

(ref. AFNOR EN 133006-2001)

2.1 **Service activities:**

They combine all activities delivered at the customer workplace by any TLD representatives including TLD, service partners:

- **Equipment Warranty Period Support:** Activities such as supervision, commissioning, project management and training sold with the equipment. These activities shall include corrections of deficiencies and any site work that might occur **during** the warrantee period of the equipment (Warranty Claim). Such activities are usually at the costs of TLD
- **Technical Assistance:** Activities occurring after hand-over of the equipment to the end-user. Such activities can be either at the specific customer request during warranty period or **after** the end of the warranty period. Such activities are usually at the costs of the customer and can be of different nature such as:
 - **Troubleshooting and service calls:** On-site or off-site activities performed after equipment failure.
 - **Audit and Expertise:** Activities such as equipment condition based audits, maintenance and operations evaluations
 - **Service Bulletin (SB):** Activities that take place to ensure maximum reactivity in the process of sending parts or field technicians to a customer during a project, breakdown or guarantee period

- **Training**: Services provided after hand-over of the equipment to the end-user at the customer request or if not sold with the equipment. Such activity is usually at the cost of the customer.
- **TKMC**: Turn-Key Maintenance Contract delivered by TLD or its agents. Any maintenance activities that are provided to the customer within the framework of a contract (see maintenance section for details).

2.2 Spare parts:

- **Capital spares**: Purchased by the customer with the main equipment. They are also called 2-year spare parts or insurance spare parts. They often are quite expensive and critical parts which are capitalized and depreciated.
- **Spare parts**: They are parts of products or systems sold separately and used by the customer to replace a corresponding faulty component.

3 Maintenance terminology and definition:

3.1 Preventive Maintenance (PM/PMI)

Preventive maintenance (PM) and Preventive Maintenance and Inspection (PMI) combine all actions taken to anticipate and avoid malfunctions and failures of an equipment whilst they are linked to either observable degradations (systematic preventive maintenance) or measurable by their effect (condition-based maintenance). They are generally planned and predetermined actions that are performed based on a yearly general planning. Usual Preventive maintenance tasks are:

- Technical cleaning
- Greasing and lubrication, filter changes
- Inspections
- Monitoring
- Vibration and oil analysis
- Partial or total technical check-up
- Repairs following inspections
- Adjustments required to avoid excessive wear

3.2 Corrective Maintenance (CM)

All actions taken following a malfunction or failure of an equipment which generates losses in production, safety or environment. Corrective actions are usually taken to restore the equipment to a workable status in a very short time such as:

- Troubleshooting done by shift support at the request of operations or following a failure requesting immediate action
- All actions performed to repair and put the equipment back to a serviceable status.

3.3 Major service:

Main objective of major service is to counter major events and failures and ensure sustainability of the operations. It is usually performed at a frequency over 1-3 year and can costs approx. 10 to 15% of routine maintenance cost

3.4 Repair:

Repair is the restoration of an item to serviceable condition through correction of a specific failure or unserviceable condition.

3.5 Overhaul:

Overhaul is the restoration of an item of an item to a standard as nearly as possible to its original condition in appearance, performance, and life expectancy.

3.6 Modernization:

Modernization has the same roots as above but addresses the issue of component obsolescence and improving functionality. It requires higher costs than a refurbishment and involved largely Engineering and design. It is sometimes called “modern life program” and his subject to Capex.

4 Maintainability and Equipment Performance Criteria:

There are several Key Performances Indicators to evaluate and benchmark the performances of GSE operations, maintenance and equipment.

4.1 Definitions of the different Operating Times:

Several operating times (up-time) can be used to measure a specific level of performance. Each operating times are impacted by a period of time during which equipment is in down status i.e. not available to perform its required function (down-time):

Operations downtimes:

- Closing time: Period of time when Airport is not operating and none of the operations and/or maintenance activities occur
- Non operating time: Period of time when the equipment is available but not required for operations. For example equipment is parked on the ramp ready to be operated but there are no aircraft to handle.
- Wait time: Period of time when equipment is available and required for handling operations but not performing its duty (engine off). For example, a loader is in place ready to load/unload the aircraft but operator is not using it (waiting for pallets or transporter...)

Maintenance downtimes:

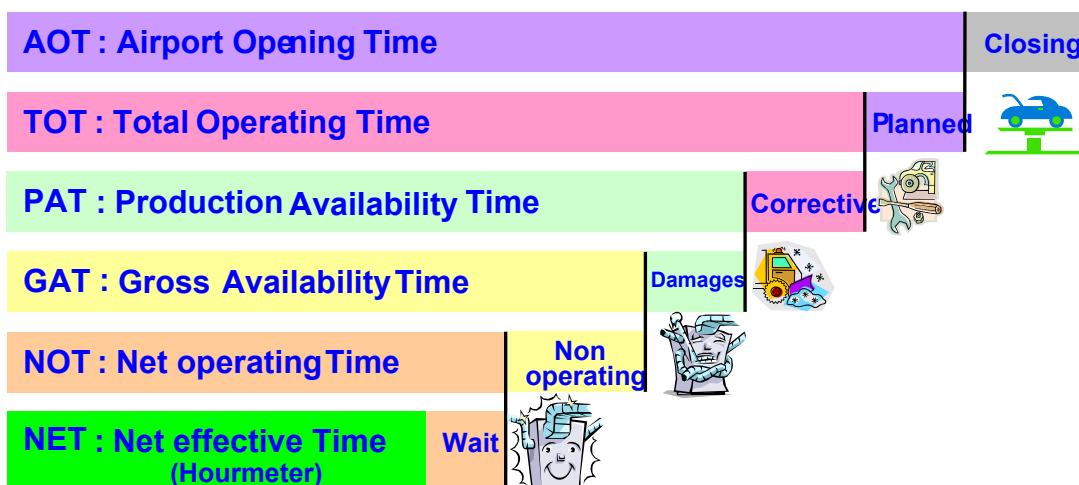
- Planned downtime: Period of time equipment is unable to perform a required function due to scheduled preventive maintenance
- Corrective Downtime: Period of time required to carry out maintenance after fault recognition with the intention to put equipment into a state in which it can perform a required function.
- Damage: Damage is an undesirable event that leads to material damage to a system, tool or machine as a direct result of incorrect use of the equipment by the operators.

Opening time:

- YOT: Yearly Opening time total number of hours in a calendar year
- AOT: Define the total number of hours Airport was open for operations

- TOT: Define the time equipment was used free of preventive maintenance.
- PAT: Define the total time equipment was used for operations free of preventive and corrective maintenance. Equipment could then be either in operations or out for damage repairs.
- GAT: Define the total time equipment was operated free of defects, breakdowns and repairs following damages. Equipment is considered to be ready for operations but may not be used to perform any activities.
- NOT: Define the total time equipment was taken for Aircraft handling operations but not performing any activities.
- NET: Define the total time equipment was effectively operated. It is typically measured by the Hour meter.

YOT : Yearly Opening Time = 24H x 365 days = 8,760 H



4.2 Reliability, maintainability, availability, MTBF and MTTR

Reliability and MTBF:

Reliability is the probability that a system will perform its intended function for a specified period of time under a given set of conditions. Reliability performance indicator can be measured either as a ratio of Production Operating Time (POT)

over the Total Operating time (TOT) or evaluate the weight of corrective downtime over the Production Operating Time.

$$\text{Equipment Reliability} = \frac{\text{PAT}}{\text{TOT}}$$

$$\text{Mean Time Between Failure (MTBF)} = \frac{\text{PAT}}{\Sigma \text{ of breakdowns}}$$

Maintainability (MTTR)

Maintainability is the probability that a given maintenance task will be done in a given timeframe and within the framework of a specified protocol. It is calculated by the sum of the time requested to perform each maintenance task divided by the number of tasks.

$$\text{Mean Time To Repair (MTTR)} = \frac{\text{PAT} - \text{TOT}}{\Sigma \text{ of breakdowns}}$$

Availability

Availability is the resultant of the maintainability and reliability factors. It refers to the equipment being available for production when scheduled

$$\text{Equipment Availability} = \frac{\text{NOT}}{\text{PAT}}$$

4.3 Equipment performances – Utilisation ratio

Equipment utilisation ratio is different to the availability because it measures the time the equipment is used (NET) divided by the time the machine was available to be used (NOT)

$$\text{Utilisation ratio} = \frac{\text{NET}}{\text{NOT}}$$

5 Total Cost of Ownership (TCO):

5.1 Introduction:

TCO is a method used to ensure that all costs related to buying, operating and maintaining an equipment for a set period of time are taken into account. It is a powerful tool to evaluate and compare accurately and consistently the financial impact of a specific equipment or fleet in the GSE operations. It is also a very comprehensive device to benchmark equipment, operations and maintenance in the industry and as such requires structure, norms and standards that will enable:

- Option evaluation when procuring new assets,
- Comparison of actual costs for similar asset types,
- Decision-making to minimize whole-life costs throughout the life of an asset,
- Feedback into future design.

5.2 Objective:

The objective of this document is to set out the structure surrounding the TCO model i.e. terminology and model.

5.3 Terminology

There are three main categories of inputs required to compute the TCO model i.e. costs, life cycle costs and expected level of performance.

5.3.1 Cost drivers:

- o Capital Expenditure (Capex):

They are non-recurrent costs and usually linked to the cost to buy and finance the equipment including:

- Cost of the equipment
- Cost of transport and custom duties
- Cost of installation and commissioning
- Cost of the capital
- Residual value i.e. estimated value of the goods at the end of the set period of time

- o Preventive maintenance (PM):

All costs related to performing PM (see definition above) including:

- Direct Labour
- Overheads
- Sub-contractors
- Mandatory inspections
- o Corrective maintenance (CM):

All costs related to performing CM including breakdowns and repairs after damages

- Direct Labour
- Overheads
- Sub-contractors
- o Operating costs:

All costs related to operating the equipment including:

- Cost of downtime and non-availability
- Labour
- Cleaning
- Petrol or other source of energy
- Wear and consumable such as tyres (refer to Spare parts consumable definition)
- o Operational expenses:

All costs related to achieve mandatory operational processes, safety and environment including:

- Environmental and safety compliance
- Airport and local authorities regulation compliance
- Training
- o Major services:

All costs related to performing major services (see definition here above)

- o Spare parts:

All costs related to the spare parts activities including:

- Cost of the spare parts
- Cost to buy spare parts
- Cost to manage spare parts logistics and inventory
- Costs of the spare parts inventory

5.3.2 Timeframe:

There are several timeframes that would need to be taken into account:

- Economic life is the estimated number of years until that item no longer represents the least expensive method of performing the functions required of it.
- Technological life is the estimated number of years until technology causes an item to become obsolete.
- Useful life is the estimated number of years during which an item will perform the functions required of it in accordance with some pre-established standard.
- Operating times: Operating time in correlation with expected level of performance is important to measure factors such as usage rate of the equipment and maintenance costs.

5.3.3 Unit of value:

There are several ways to present the outputs of a TCO mainly according to customer value. As such, it can be either a straight calculation in terms of financial cost over a period of time (e.g. \$/5 years) or hours of operations (\$/H). In some cases, customer might value cost per operation leading to unit of measure such as for example, total cost to load a pallet or by KWH produced...

5.3.4 Which TCO?

TCO can be presented at three different levels:

- TCO 1: limited to cost to purchase the goods and maintain it.
- TCO 2: TCO 1 + costs to operate (excluding operator related costs)
- TCO 3: TCO 2 and costs linked to operators

5.3.5 Model:

The following fundamental concepts are common to all applications of TCO:

- Cost breakdown structure:(see above)

Model will have to be structured in such a way as to allow analysis of specific areas. For example, the purchaser might need to compare spares costs for each option; these costs should therefore be identified within the structure i.e. TCO1, 2 and 3

- Economic appraisal:

We shall provide customers with a model that will enable comparisons and benchmarking. Several financial aspects will need to be included:

- Cost-benefits analysis: compare costs and benefits that occur in different time periods.
- Net present value: As the present is the most suitable time reference, all future costs must be adjusted to their present Value using a discounted rate.

Discount rates used by the GSE industry could vary considerably and care must be taken when comparing TCO analyses which are commercially prepared to ensure a common discount rate is used.

- Residual value: Residual value should correspond to the value we could expect to sale the equipment for. Standard practice applied to our TCO model shall be:

Category of equipment	Equipment concerned	5 years	10 years	15 years
Type 1	Conventional tractors, Towbarless Tractors and Loaders:	40%	25%	15%
Type 2	GPU, ASU, ACU, TF:	30%	20%	10%
Type 3	NBL, ABS, EBT, LSP, WSP	15%	10%	5%